stable-diffusion-v1-5-inpainting

Beta

Text-to-Image • runwayml

`@cf/runwayml/stable-diffusion-v1-5-inpainting`

Stable Diffusion Inpainting is a latent text-to-image diffusion model capable of generating photo-realistic images given any text input, with the extra capability of inpainting the pictures by using a mask.

Model Info

Terms and License	link ↗
More information	link ↗
Beta	Yes

Model Info

Unit Pricing	$0.00 per step

Usage

Workers - TypeScript

TypeScript

```
export interface Env {
  AI: Ai;
}

export default {
  async fetch(request, env): Promise<Response>
{

    // Picture of a dog
    const exampleInputImage = await fetch(
      "https://pub-
1fb693cb11cc46b2b2f656f51e015a2c.r2.dev/dog.png
    );

    // Mask of dog
    const exampleMask = await fetch(
      "https://pub-
1fb693cb11cc46b2b2f656f51e015a2c.r2.dev/dog-
mask.png"
    );

    const inputs = {
      prompt: "Change to a lion",
```

```
      image: [...new Uint8Array(await
exampleInputImage.arrayBuffer())],
      mask: [...new Uint8Array(await
exampleMask.arrayBuffer())],
    };

    const response =
      await env.AI.run(
        "@cf/runwayml/stable-diffusion-v1-5-
inpainting",
        inputs
      );

    return new Response(response, {
      headers: {
        "content-type": "image/png",
      },
    });
  },
} satisfies ExportedHandler<Env>;
```

curl

```
curl
https://api.cloudflare.com/client/v4/accounts/$
diffusion-v1-5-inpainting  \
  -X POST  \
  -H "Authorization: Bearer $CLOUDFLARE_API_TOI
  -d '{ "prompt": "cyberpunk cat" }'
```

Parameters

* indicates a required field

Input

- `prompt` **string** required min 1

  A text description of the image you want to generate

- `negative_prompt` **string**

  Text describing elements to avoid in the generated image

- `height` **integer** min 256 max 2048

  The height of the generated image in pixels

- `width` **integer** min 256 max 2048

  The width of the generated image in pixels

- `image` **array**

  For use with img2img tasks. An array of integers that represent the image data constrained to 8-bit unsigned integer values

  - `items` **number**

    A value between 0 and 255

- `image_b64` **string**

  For use with img2img tasks. A base64-encoded string of the input image

- `mask` **array**

  An array representing An array of integers that represent mask image data for inpainting constrained to 8-bit unsigned integer values

- `items` **number**

  A value between 0 and 255

- `num_steps` **integer** default 20 max 20

  The number of diffusion steps; higher values can improve quality but take longer

- `strength` **number** default 1

  A value between 0 and 1 indicating how strongly to apply the transformation during img2img tasks; lower values make the output closer to the input image

- `guidance` **number** default 7.5

  Controls how closely the generated image should adhere to the prompt; higher values make the image more aligned with the prompt

- `seed` **integer**

  Random seed for reproducibility of the image generation

Output

The binding returns a `ReadableStream` with the image in JPEG or PNG format (check the model's output schema).

API Schemas

The following schemas are based on JSON Schema

Input Output

```
{
    "type": "object",
    "properties": {
        "prompt": {
            "type": "string",
            "minLength": 1,
            "description": "A text description of the image you want to generate"
        },
        "negative_prompt": {
            "type": "string",
            "description": "Text describing elements to avoid in the generated image"
        },
        "height": {
            "type": "integer",
            "minimum": 256,
            "maximum": 2048,
            "description": "The height of the generated image in pixels"
        },
        "width": {
            "type": "integer",
            "minimum": 256,
```

```
            "maximum": 2048,
            "description": "The width of the
generated image in pixels"
        },
        "image": {
            "type": "array",
            "description": "For use with
img2img tasks. An array of integers that
represent the image data constrained to 8-bit
unsigned integer values",
            "items": {
                "type": "number",
                "description": "A value
between 0 and 255"
            }
        },
        "image_b64": {
            "type": "string",
            "description": "For use with
img2img tasks. A base64-encoded string of the
input image"
        },
        "mask": {
            "type": "array",
```

```
            "description": "An array
representing An array of integers that
represent mask image data for inpainting
constrained to 8-bit unsigned integer values",
            "items": {
                "type": "number",
                "description": "A value
between 0 and 255"
            }
        },
        "num_steps": {
            "type": "integer",
            "default": 20,
            "maximum": 20,
            "description": "The number of
diffusion steps; higher values can improve
quality but take longer"
        },
        "strength": {
            "type": "number",
            "default": 1,
            "description": "A value between 0
and 1 indicating how strongly to apply the
transformation during img2img tasks; lower
```

```
values make the output closer to the input
image"
        },
        "guidance": {
            "type": "number",
            "default": 7.5,
            "description": "Controls how
closely the generated image should adhere to
the prompt; higher values make the image more
aligned with the prompt"
        },
        "seed": {
            "type": "integer",
            "description": "Random seed for
reproducibility of the image generation"
        }
    },
    "required": [
        "prompt"
    ]
}
```

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